

X-Ray Device Positioning with Augmented Reality Visual Feedback

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ABSTRACT

In minimally invasive surgeries one common way to verify progress is the use of an intraoperative X-ray device (due to its characteristic shape called a C-arm). Its control, however, remains challenging owing to its complex movements. We propose the use of an Augmented Reality Head-Mounted Display (AR-HMD) to let the surgeon choose a desired X-ray view interventionally providing the corresponding C-arm configuration as visual feedback. The study participants' feedback, despite being critical of the HMD hardware limitations, suggests an inclination towards using AR for orthopaedic surgeries on especially complex or unusual anatomies.

Index Terms: Computing Methodologies—Computer graphics—Graphics Systems and Interfaces—Mixed / Augmented Reality; Human-Centered Computing—Human Computer Interaction (HCI)—Interaction Paradigms—Mixed / Augmented Reality

1 INTRODUCTION

The trial-and-error methodology of positioning a C-arm to obtain specific intraoperative X-rays for monitoring the progress of minimally invasive surgeries increases the radiation exposure to the patient and surgical staff along with surgery duration. One approach to simplify the finding of the optimal pose of the C-arm is to track the X-ray device and generate synthetic X-rays, also known as Digitally Reconstructed Radiographs (DRR), from this pose dynamically [1]. While intuitive, this is time-consuming and cumbersome. Another project describes a system to re-position a C-arm into its previous pose by visualizing the recorded C-arm pose [5].

Desired-view controlled positioning [2] handles the complexities of C-arm positioning to acquire a specific view of the anatomy by generating DRRs from patient data in pre-surgery planning. We circumvent the need for a pre-operative planning stage by combining intraoperative planning with simultaneous visualization.

2 METHODS

We present an AR approach to position the C-arm based on a Computed Tomography (CT) scan of the patient undergoing surgery. The application enables the surgeon to select a simulated X-ray projection as a view on the anatomy. This view relates to the position of the C-arm through patient to CT registration.

Our implementation was done for Microsoft HoloLens 1, using Unity 3D with marker tracking by Vuforia, and Microsoft Mixed-Reality Toolkit to provide gesture interaction.

The positioning of the C-arm based on a DRR view is described by forward kinematics: in the application the surgeon adjusts the individual joints of the C-arm, which directly leads to an updated perspective on the patient anatomy. The relation between the pose of the C-arm (its position, angulation, orbital rotation etc.) and



Figure 1: The AR UI in front of a real C-arm. The surgeon can select a real-time DRR view on the anatomy, by moving the sliders.

the DRR are calculated using the Denavit-Hartenberg rules. These DRRs are rendered from the preoperative patient CT data using volume raycasting implemented by appropriate shaders. With the DRR now available for a given pose of C-arm with respect to the patient volume, the next step is to develop a control mechanism to manipulate the virtual C-arm.

A touch-less User Interface (UI) (Fig. 1) is provided, which is a viable method of interaction preserving the sterility of the surgeons hands [3]. An HMD can project the UI, which enables the surgeon to both view the DRRs and interact with it through hand gestures. The projection of the UI floating above the patient was done to accurately present the DRR, minimize obstruction, and be conveniently visible to the surgeon. The UI allows the surgeon to control, using pinching slider gestures, the cranial/caudal angulation, left/right orbital and superior/inferior translation movements of the C-arm with respect to the patient. Once the surgeon has arrived at the desired view, it can be saved for retrieval later. The surgeon can switch to a virtual representation of the C-arm (Fig. 2) to visualize it in the surgical environment, which can assist in its positioning.

The surgeon can check the visualization of the suggested C-arm pose for feasibility, collision-free placement in the operating room, and ease of access to the patient. Such subjective surgical environment criteria naturally cannot be assessed automatically, but must be verified by a medical expert reviewing the visualization.

3 EXPERIMENT

A user study was conducted to establish the face validity and usability of the developed application contrasted to the trial and error C-arm positioning method. It indicates acceptance, comfort and potential of AR frameworks in the surgical domain.

3.1 Study Procedure

After filling in a pre-study questionnaire (demographic information, estimates of their media usage, clinical experience and technology

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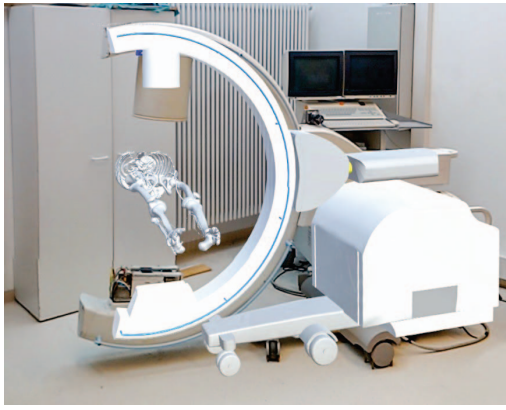


Figure 2: The selected view corresponds to a C-arm pose in relationship to the patient and environment. We show the virtual C-arm in this pose which can guide the surgeon to position the real C-arm.

acceptance [4]), we let the participants experience the system for several minutes exploring all functionalities of the application. After exposure to the application, we had them fill in questionnaires regarding usability and task load and record their experiences.

We assessed the perceived usability by including the System Usability Scale (SUS) using the common 5-point scale (1 - strongly disagree, 5 - strongly agree). As another measure, we assessed the perceived task load using the NASA Task Load Index (TLX), and report on its individual subscales. Finally, we asked free-text questions for advantages and disadvantages of the method.

3.2 Participants

The study included $N = 4$ participants ($M_{age} = 29.0$, $SD_{age} = 2.2$) from the medical field. Three of them were assistant physicians and one of them a final-year medical student. Participants stated to spend time with digital media (PC, mobile phone, etc.) for about 40.8 hours per week ($SD = 20.1$). Three participants noted to have used Mixed Reality systems before. Participants stated to perform on average 2.6 image guided surgeries per week ($SD = 1.9$). Their technology acceptance facet according to the technology commitment questionnaire is very high (average 19.5 on a scale of 0 – 20.0).

4 RESULTS

The SUS resulted in an average of $M_{SUS} = 70.625$ ($SD_{SUS} = 8.75$), which puts the system into a “good” range. The accumulated individual subscales of the NASA-TLX are reported in Fig. 3. All the participants found the HMD to be uncomfortable for extended use in

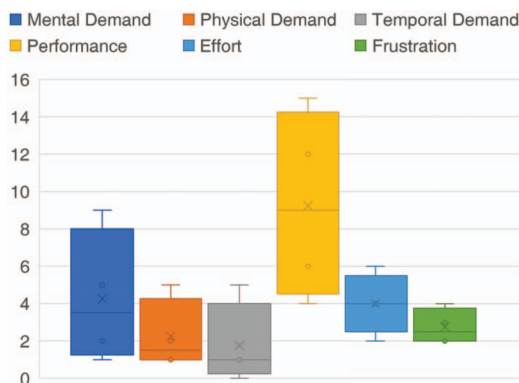


Figure 3: Individual subscales of the NASA-TLX questionnaire.

surgery, and most of them complained about the field of view of the HoloLens 1 to be limited. The participants found that the application would be most advantageous for orthopaedic and trauma surgery.

5 DISCUSSION

The participants valued the AR visualization of the patient and the 3D display of the C-arm highly. One participant wished for inclusion of the application in even more modalities. However, the inconvenience of wearing a heavy device was stated as a deterrent to the acceptance of HMDs in surgery. For experienced surgeons under normal surgical conditions, some trial and error X-ray images are more appealing than the use of cumbersome devices. They were however inclined towards using AR for complex surgeries, such as shoulder, spine, and ankle surgery. Also AR was reported to be favourable when dealing with patients with physical deformities, and other factors that made the positioning of a C-arm unintuitive.

One participant noted that the larger field of view of the DRR compared to an actual C-arm, helped to make the positioning easier. In this regard, the simulation does not need to be constrained by the capabilities of the physical X-ray device, as long as it provides a modality that the users are familiar with. This salient feature of the application can prove beneficial for training of medical students in standard views without exposing them to radiation.

One participant raised the concern, that the rest of the surgical team is excluded from the AR application, as they are not able to see the surgeon’s augmented view. Therefore multi user support would be a desirable improvement. Another participant suggested natural control of the C-arm through a virtual grabbing interaction. Further two participants requested to see both the DRR and the C-arm in its corresponding pose simultaneously as opposed to individually.

Future improvements of the UI could include real-time X-rays of the C-arm to show angiographies during surgery, and automation of C-arm movement according to the chosen position.

6 CONCLUSIONS

In this paper we presented the use of an HMD application as an intraoperative planning tool for C-arm positioning. In our study we found that the participants generally liked the idea of visualizing the 3D representation of the patient and the C-arm in a consistent way and the possibility to receive live updates of the DRR relative to the C-arm pose. Barring the discomfort of wearing an HMD for long duration, the inclusion of an HMD into image guided surgery with its touch-less user interface and powerful visualization methods can increase the efficiency and quality of medical procedures.

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